

Health Disparities in Nephrology

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KEY POINTS

- In the United States, significantly lower incidence of end-stage kidney disease (ESKD) and smaller health insurance–related gaps in care access were recently observed in states with broader as compared with narrower health insurance coverage for the poor.
- Targeted efforts by the US Indian Health Service to improve screening and management of diabetes mellitus and hypertension have led to a 54% decline in the incidence of diabetes-related ESKD among American Indians over the past 2 decades: the age-adjusted incidence rate of diabetes-related ESKD is now the same as that for non-Hispanic white Americans.
- Emerging evidence suggests that the dialysis center may play an influential role in promoting and facilitating access to kidney transplant, particularly among racial–ethnic minority patients.
- Gap programs aimed at increasing kidney donation in the United States particularly among black Americans have yielded a 30% increase in the donation rate of deceased-donor kidneys (from 18.1 per 1000 deaths in 2000 to 23.6 per 1000 deaths in 2014) from this racial group.
- In the early months following implementation of a new Kidney Allocation System, incidence rates of deceased-donor kidney transplantation improved by 13% for Hispanic and 19% for black patients in the United States.
- The black–white disparity in renal allograft loss after both deceased- and living-donor kidney transplantation has steadily declined over the past 2 decades due to differential improvements in 5-year graft survival among black as compared with white Americans.

Disparities in the incidence, progression, and treatment of kidney diseases according to socioeconomic status, race–ethnicity, and geographic location have been documented for many years. These relative differences arise and persist via multiple intermediaries, including biologic susceptibility, differential access to high-quality health care, and contextual impediments to healthy living. This chapter provides a framework for evaluating systematic differences in health, describes what is known about root causes of disparities in nephrology

and nephrology care, and highlights broad strategies for addressing key determinants of these inequities.

DISPARITY

The term “health disparity” often refers to suboptimal health processes or outcomes experienced by demographically defined groups that occur in the context of social or economic

inequality. The United States National Library of Medicine defines a “health status disparity” as variation in rates of disease occurrence and disabilities between (among) socioeconomic and/or geographically defined population groups.¹ By contrast, a “health-care disparity” refers to a difference in access to or availability of health-care facilities and services.¹ Health status and health-care disparities relate to systematic inequalities in health among social groups that are deemed to be preventable by reasonable means.

This chapter covers five broad areas of health status and health-care disparities in nephrology observed in socially disadvantaged groups: 1. disparities in the incidence and prevalence of risk factors for chronic kidney diseases (CKDs); 2. disparities in the treatment of these risk factors and in progression of CKD; 3. disparities in the incidence of end-stage kidney disease (ESKD); 4. disparities in the treatment of ESKD; and 5. initiatives to reduce these disparities.

Global estimates of ESKD incidence vary widely and are strongly influenced by the availability of funding for the treatment of kidney diseases and associated risk factors.² Worldwide approximately 2.6 million people received treatment for ESKD in 2010; however, estimates suggest that roughly 2.3 to 6.1 million people may have died from ESKD because renal replacement therapy was declined or not accessed.³ In fact, prior to passage of US Public Law 92-603 in 1972, access to maintenance dialysis in the United States was commonly based on the candidate’s financial health and social worth criteria such as occupation.⁴ Such biases remain operative in many countries, where people of low socioeconomic status and members of socially disadvantaged minority groups experience disproportionately high incidence of ESKD.^{5–8} For example, in the United States, approximately 25% of black Americans reside in poverty as compared with less than 10% of non-Hispanic white Americans.⁹ Black Americans also experience disproportionately high rates of ESKD,^{2,7} and yet for over 2 decades they have been consistently less likely than their white counterparts to receive a living- or deceased-donor kidney transplant.^{2,10,11} Although much less is known about people with kidney diseases who do not require or pursue dialysis, reports suggest that compared with US adults with earlier stages of CKD, those with more advanced CKD are younger and more likely to be nonwhite, poor, and uninsured.^{2,12} Similar disparities in the incidence, treatment, and risk factors of ESKD have been observed among indigenous people from North America, Australia, and New Zealand.^{13–21} The magnitude and persistence of these disparities have prompted some governments to prioritize elimination of the disparities while also attempting to reduce the overall burden and costs attributed to kidney diseases.^{22,23}

HEALTH JUSTICE

In 1966, Dr. Martin Luther King, Jr. proclaimed, “Of all the forms of inequality, injustice in health care is the most shocking and inhumane.” The pursuit of equal health outcomes or equal access to health care plays a prominent role in contemporary debates on health care.^{24,25} Examples of equal access and equal outcomes with respect to health care reflect the Aristotelian principles of horizontal and vertical equity.²⁶ However, such debates in health care often include conflicting views of justice. For example, a medical model of justice

focuses attention at a point in time when someone who is already ill seeks access to scarce and/or an expensive service such as dialysis or kidney transplantation. (Many credit the birth of bioethics in the United States to the introduction of the arteriovenous shunt by Belding Scribner in 1960 which allowed for long-term hemodialysis to treat patients with ESKD.) By contrast, a social model of justice highlights how social determinants such as access to education, employment, preventative care, safe neighborhoods, and nutritional foods affect the health of a population.^{25,27} In other words, the social model focuses on how a person’s need for health care arose in the first place. Although inequities in distributive justice (e.g., in kidney transplantation) persist in many areas of medicine, health status and health-care disparities in nephrology have focused growing attention on the link between social conditions and risk factors that place people at risk for progressive diseases of the kidney such as low birth weight, obesity, type 2 diabetes mellitus, and hypertension.^{24,28}

SOCIAL ORIGINS OF DISPARITIES IN NEPHROLOGY

Link and Phelan originally referred to social conditions, including social class, race, income, and education level, as “fundamental causes of disease because these conditions govern access to resources that influence health and disease.”²⁹ Prior studies have reported that the incidence of progressive CKD and ESKD is highest among persons living in the most impoverished neighborhoods.^{6,30,31} Additional US-based studies have reported lack of health insurance coverage and residence in a high-poverty area as exposures linked with worse kidney-related biochemical abnormalities at dialysis initiation and marked delays in receipt of a kidney transplant.^{32–34} Although these disparities arise from a variety of mechanisms, including biologic susceptibility and economic vulnerability, environmental exposures such as contextual impediments to healthy living likely play a central role. Discussion of the hypothesized influences of social conditions on disparities in nephrology through various life stages uses the framework shown in Fig. 83.1.

PRENATAL ENVIRONMENT AND DISPARITIES IN RISK FACTORS FOR KIDNEY DISEASES

In 1990, Barker noted, “The womb may be more important than the home” in terms of the origins of adult disease.³⁵ Worldwide, every year approximately 15 million newborns (10 per 100 live births) are born prematurely and 22 million (16 per 100) weigh <2500 g (i.e., low birth weight).^{36,37} While intrauterine growth retardation and low birth weight are invariably linked with measures of maternal poverty,^{38–40} much debate surrounds the influence of early fetal programming on subsequent risk of chronic diseases including those of the kidney (see Chapter 21 for a detailed discussion). The “critical periods” model is one of three models that has been proposed to explain how socioeconomic position over the life course may influence subsequent health.⁴¹ The “fetal programming” hypothesis is based on the critical periods model and states that intrauterine growth retardation results in permanent alterations in fetal physiology that become deleterious later in adult life.³⁵ Brenner hypothesized that

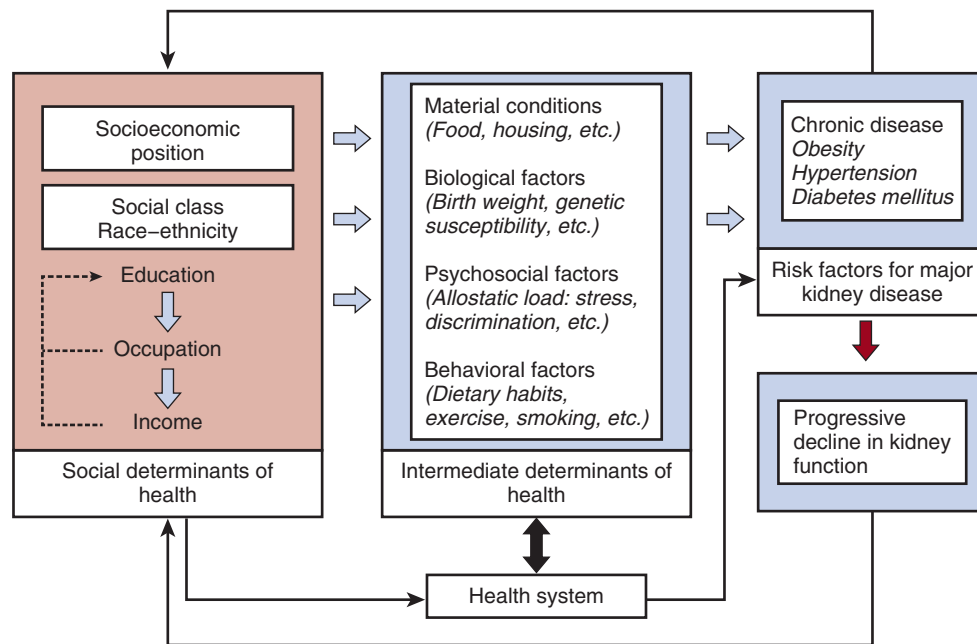


Fig. 83.1 Social determinants of health disparities framework.

retardation of renal development as occurs in individuals of low birth weight both increases postnatal risks for systemic and glomerular hypertension and predisposes an individual to further nephron loss and the development of progressive kidney injury.^{42–44} This hypothesis draws support from reports of 1. a direct relationship between birth weight and nephron number,⁴⁵ 2. an inverse relation between birth weight and hypertension later in life,^{35,46–48} and 3. an inverse relation between nephron number and blood pressure.⁴⁹

In addition to hypertension, low birth weight has been variably linked with risk factors such as diabetes and obesity and has been directly associated with later-life onset of microalbuminuria, reduced estimated glomerular filtration rate (eGFR), and ESKD.^{50–53} One study conducted on a cohort of term singletons born around the time of the 1944–1945 Dutch famine found that midgestational exposure to famine was associated with microalbuminuria and that late gestational exposure was associated with glucose intolerance.⁵¹ Notably, a rapid increase in nephron number occurs in midgestation, which is a critical period for determining nephron endowment at birth.⁵¹ Moreover, rapid growth and weight gain during infancy and childhood that occur after intrauterine growth restriction (and/or low birth weight) are associated with increased risks of insulin resistance, obesity, diabetes, and hypertension.^{42,53,54}

Because low birth weight is strongly linked to indices of parental poverty and occurs more commonly among socially or economically disadvantaged groups, researchers have posited that low birth weight may partly account for socioeconomic and racial-ethnic disparities in congenital, hypertensive, and hyperfiltrative kidney diseases.^{55,56} For example, Australian Aboriginals experience higher rates of poverty and an approximate tenfold higher risk of ESKD than nonindigenous Australians.⁸ One study conducted in a remote Australian Aboriginal community (where low-birth-

weight infants compose approximately 25% of live births) observed that low as compared with high birth weight was significantly associated with lower kidney volumes, higher blood pressure, and later in life, higher levels of urinary albumin excretion.⁴⁹ In a recent study of 447 Japanese children, low as compared with normal birth weight was associated with a nearly fivefold higher risk of having moderate or severe elevations in serum creatinine levels during childhood (mostly due to congenital anomalies of the kidney and urinary tract).⁵⁷ In the Bogalusa Heart Study, a biracial prospective cohort study of cardiovascular risk factors in children and adolescents in the Southern United States, the prevalence of low birth weight was higher among black than white infants.⁵⁸ Subsequent reports have linked lower birth weight in this cohort with higher blood pressure levels later in life. These reports have also noted that the magnitude of the association between lower birth weight and higher blood pressure appears to increase with advancing age.^{46,47,58}

Globally, many high- and medium-income countries now face rising incidence of maternal overnutrition.^{59–61} In some countries, maternal obesity has replaced tobacco smoking as the most important preventable risk factor for adverse pregnancy outcomes including preterm births.^{59,61} In addition to preterm delivery, excessive weight gain during pregnancy is associated with increased risk for development of obesity, diabetes, and hypertension in the offspring.⁵⁹ Less is known, however, about the pathways through which excessive weight gain during pregnancy and maternal obesity affect later-life CKD risk factors. The degree to which maternal obesity and excessive weight gain during pregnancy contribute to socioeconomic and racial-ethnic disparities in kidney disease incidence is likewise unknown. The trend toward increasing survival of low-birth-weight infants further underlines the importance of intrauterine fetal environment and timing of insults in future risk for development of precursors to progressive kidney disease.⁵⁵

POSTNATAL ENVIRONMENT AND DISPARITIES IN RISK FACTORS FOR KIDNEY DISEASES

Over the past 2 decades, the nephrology community has paid increasing attention to how a person's living environment may influence his or her downstream risk of chronic diseases. This heightened attention is evidenced by the growing body of literature examining the complex influences of individual- and area-level measures on important processes and outcomes related to the care of people with CKDs.^{31,62–64} People residing in impoverished neighborhoods experience more crime, less access to high-quality educational and employment opportunities, and poorer health outcomes in comparison with persons from less distressed areas.^{63,65} Several contemporary studies have described higher incidence of ESKD as well as higher prevalence of albuminuria and reduced eGFR in areas of high residential poverty than in areas of low residential poverty.^{31,62,64} However, much debate surrounds how and how much adverse social conditions in childhood and early adult life influence a person's subsequent risk of developing chronic diseases of the kidney.²⁸ Many writers posit that social conditions act through a host of intermediaries such as education, housing, and diet, thereby molding health-related behaviors, which in turn influence risk for kidney diseases.^{28,29,63} Prior studies suggest that education may influence disparities in health behaviors through its effects on lifestyle, social support, and social norms, whereas income may drive disparities in health outcomes through access to resources and services, including health care.^{66,67} The social distribution of important risk factors for progressive diseases of the kidney, including the higher prevalence of smoking, low birth weight, type 2 diabetes mellitus, hypertension, and obesity within poor as compared with less impoverished areas, lends evidence to these claims.^{24,28} This unequal distribution of intermediary exposures linked to progressive kidney injury is further compounded by the vulnerability of certain groups (i.e., persons of low socioeconomic position) to health-compromising illnesses such as ESKD. Poor health then exerts potent feedback on an individual's socioeconomic position through lost income or medical costs associated with a major illness (Fig. 83.1).⁶⁸

The concept of “allostatic load,” the accumulation of physiologic insults due to repeated insults or chronic stressors experienced in daily life, has gained increasing interest as a model for explaining socioeconomic (and racial-ethnic) disparities in the processes and outcomes of chronic diseases.^{69,70} Specific measures of allostatic load include levels of hormones secreted in response to stress and biomarkers that reflect the effects of these hormones on the body.^{69,71} This concept, which is based on the cumulative exposures theory, posits that stress may accumulate from early life and manifest as an accumulation of physiologic dysregulation.^{69,72} Chronic stressors such as food insecurity, discrimination, living in substandard housing, inadequate access to health care, and exposure to violence are more pronounced among persons of lower socioeconomic status than among those of intermediate or higher socioeconomic status.^{70,71} Such psychosocial stressors may lead to hypertension and autonomic dysregulation (e.g., diminished heart rate variability), which in turn are linked to increased risk of incident ESKD.^{70,71,73} Accordingly, higher levels of exposure and adaptation to life stressors experienced more commonly by the poor or

members of some racial-ethnic minority groups may put them at higher risk for kidney disease through the aforementioned pathways than their less impoverished or white counterparts.^{74–76}

Similarly, contemporary debates in many industrialized countries have increasingly focused on the role of nutrition and childhood obesity in later-life risks for development of chronic disease.^{77,78} Estimates suggest that more than a third of children in North America and Europe, and approximately a quarter in the western Pacific and southeast Asia, are overweight or obese.^{79,80} In the United States, the prevalence of childhood obesity is markedly higher among black and Hispanic children than among non-Hispanic white children.⁷⁷ Public health policy and nutrition experts attribute the marked growth in childhood obesity to the widespread availability of energy-dense, inexpensive foods and the low energy requirements of modern daily life.⁷⁹

Several studies have also linked residence in impoverished areas with higher levels of processed food consumption (rich in sodium and phosphorus), lower levels of physical activity, and increased risk for chronic diseases such as cardiovascular disease, diabetes, and hypertension in comparison with residence in more affluent areas.^{63,81–84} In addition to offering fewer options for affordable, nutritious foods, poorer neighborhoods tend to have fewer (or less attractive) areas to promote physical activity such as parks and community centers.^{83,84} The undue influence of crime, poor housing quality, and suboptimal educational and medical infrastructure may also render poor communities less supportive of healthy lifestyles than less distressed neighborhoods.⁶⁵ One study linked racial-ethnic disparities in childhood obesity with differential exposure to risk factors during early life, such as gestational diabetes, rapid infant weight gain, excessive television viewing, and high intake of sugar-sweetened beverages or fast food.⁷⁸ The strong relation between childhood obesity and later-life antecedents of hypertensive and hyperfiltrative kidney diseases (obesity, hypertension, metabolic syndrome, type 2 diabetes mellitus) suggests that interventions targeting the early life period may contribute substantially to reducing socioeconomic and racial-ethnic disparities in kidney and other chronic diseases.⁸⁵

Perhaps the strongest evidence that neighborhood socioeconomic distress may influence downstream risk factors and lead to health status disparities in certain types of kidney disease emanates from the Moving to Opportunity Study.^{86,87} During the period 1994–1998, the US Department of Housing and Urban Development conducted this “randomized social experiment” in five cities (Baltimore, Boston, Chicago, Los Angeles, and New York City) to examine the extent to which relocating from a more to a less distressed neighborhood might influence downstream health outcomes. Collectively, 4498 women (90% were either black or Hispanic) with children from high-poverty neighborhoods were randomly assigned to one of three groups: 1788 were assigned to receive housing vouchers that were redeemable only if they relocated to a low-poverty area, 1312 received unrestricted vouchers, and 1398 were assigned to a control group that offered neither of these opportunities. In terms of risk factors for kidney diseases, the study found that participants who received a voucher to relocate to a low-poverty neighborhood experienced absolute risk reductions of 5% and 4% for extreme obesity and diabetes mellitus, respectively, 10 to 15 years

later in comparison with controls.⁸⁷ Notably, the effects of relocation occurred without detectable effects on individual-level income.⁸⁸ In other words, an individual may experience substantial benefits by moving from a more to a less impoverished neighborhood in terms of reducing long-term risk for development of progressive CKD. However, the strong correlation between neighborhood characteristics (e.g., crime, educational and employment opportunities, recreation, social norms) renders the evaluation of specific pathways to these outcomes elusive.²⁷ Although provocative, the results of the Moving to Opportunity Study highlight the challenges faced by policy makers and clinicians in addressing more proximal mediators of socioeconomic disparities in important CKD risk factors. These findings also raise questions about how we might improve a person's or family's health through altering social support via simple, low-cost educational means, such as text messaging and Web-based networks.

RACE, SOCIAL CONDITIONS, AND DISPARITIES IN INCIDENT KIDNEY DISEASES

Regrettably in many parts of the world, race is highly correlated with socioeconomic conditions. Link and Phelan²⁹ emphasize that race is itself so closely tied to social and economic resources that it should also be considered a fundamental cause of health inequality. Insight into the origins of racial-ethnic disparities in kidney diseases, however, requires a firm understanding of the social context in which the inequities occur. For example, the United States has witnessed substantially higher incidence rates of ESKD among members of racial-ethnic minority groups, most prominently among black Americans, than among white Americans.^{2,7,89} Black Americans are also much more likely than white Americans to be members of the urban “underclass” (residing in areas of intensely concentrated poverty) and to lack health insurance coverage and access to a usual source of care.⁹⁰ These data led to claims that social disadvantage largely explains the disparity in risk for progressive kidney diseases between black and white Americans through pathways previously described.⁹⁰ For example, among 603 children (median age 10 years) with nonglomerular kidney diseases enrolled in the Chronic Kidney Disease in Children study, the significantly shorter time to ESKD onset among black as compared with non-black children (median difference 3.2 years) was attributable to indices of socioeconomic status.⁹¹ In adults, nearly three-quarters of the excess risk of incident albuminuria or reduced eGFR among blacks in comparison with whites in the Atherosclerosis Risk in Communities Study was likewise attributable to measures of socioeconomic status and health-care access.^{92,93} In the general US population, health-related behaviors (e.g., smoking, physical activity, alcohol intake), comorbid risk factors (e.g., diabetes mellitus, hypertension, obesity), and measures of health-care access (e.g., health insurance, routine care visits) appear to mediate varying fractions of the associations between income and the presence of albuminuria and/or reduced eGFR, accounting for 20%, 32%, and 11%, respectively, of these associations.⁹⁴ However, studies in other US health-care settings indicate that a substantial fraction of the excess risk of incident ESKD among black and white Americans remains unexplained even after one has accounted for differences in the distribution of ESKD risk factors, measures of socioeconomic status, and access to

health care.^{89,95,96} The persistence of marked racial disparities in ESKD risk within settings where socioeconomic gradients are theoretically reduced, along with emerging genetic data (see later) linking African ancestry to certain forms of progressive kidney disease, further highlights the complex influences of gene-environment interactions on these disparities.⁹⁶⁻⁹⁸

Our understanding of disparities in kidney diseases among other racial-ethnic groups is less well developed, but the origins of these disparities have common social constructs. In North America, for example, the prevalence of ESKD is substantially higher among Native Americans (primarily among American Indians and Aboriginal Canadians) than among non-Hispanic whites.^{2,20,21,99} This situation is mirrored in the indigenous people (Aboriginals, Maori, and other Pacific Islanders) of Australia and New Zealand.⁸ Differences in social circumstances and important risk factors such as obesity, diabetes mellitus, and hypertension account for substantial fractions of this excess risk of ESKD among indigenous people.^{8,17,99} In fact, long-term cohort studies of Aboriginal Australians suggest that the lifetime risk of ESKD among people with diabetes may be as high as 41% in this group.¹⁵ Moreover, the remote location of many indigenous communities combined with cultural differences in their approach to illness often hinders the implementation of interventions to adequately address these risk factors.^{14,17,19,100,101} Notably, the US Indian Health Services began systematic efforts in the 1990s to identify, screen for, and address risk factors, primarily blood pressure and diabetes control, among American Indians (see later).¹⁸⁻²⁰ These efforts have been widely touted as the primary reason for the steady decline in ESKD incidence among American Indians over the ensuing 2 decades.^{2,18} Lastly, detection of racial-ethnic disparities in nephrology strongly depends on the manner in which health statistics are collected and organized. Unlike in many European countries, where health statistics are commonly organized according to occupational or class hierarchies, health data in North America, Australia, and New Zealand have historically included race or ethnicity.

BIOLOGIC ORIGINS OF DISPARITIES IN NEPHROLOGY

Although differences in the built environment undoubtedly play a central role in promoting racial-ethnic disparities in the incidence and treatment of kidney diseases, epidemiologic and genetic studies support the presence of strong biologic influences as well.^{98,102} Notably, the higher ESKD incidence experienced by members of racial-ethnic minority groups in comparison with non-Hispanic whites exists even within integrated health systems (with theoretically equal care access) and among socioeconomically disadvantaged populations (in which socioeconomic gradients are attenuated).^{96,97,103} The persistence of racial-ethnic differences in ESKD risk in these studies suggests that factors other than socioeconomic status and health-care access play a more influential role in determining disease progression than previously inferred.^{98,102}

Certain types of kidney diseases are hypothesized to occur among susceptible individuals through a complex interaction of genetic predisposition and environmental exposures.^{98,102} In the United States, for example, blacks are twice as likely as whites to have a first-degree relative with ESKD.¹⁰⁴

Non-Hispanic black children with ESKD also have a substantially higher prevalence (51%) of nondiabetic glomerular disease than Hispanic (39%) or non-Hispanic white (35%) children. Shared exposures from the familial environment such as poverty, nutrition, and health-related behaviors may contribute to these risks, but further studies have linked *APOL1* gene mutations with progressive kidney diseases such as focal segmental glomerulosclerosis and human immunodeficiency virus (HIV) nephropathy.^{98,105} Owing partly to the protective effects conferred by *APOL1* mutations against trypanosomal disease and other pathogens, *APOL1* mutations appear to be relatively common among individuals of West African descent but virtually absent among persons from Europe, East Africa, and other regions.¹⁰⁵ It has been estimated that as much as 40% of the increased burden of progressive nondiabetic glomerular diseases in Sub-Saharan African ancestry populations could be eliminated if the injury mechanism for *APOL1* risk variants could be overcome.^{106–108} However, the detailed mechanisms whereby *APOL1* variants induce kidney injury are yet to be fully elucidated.¹⁰⁹ Furthermore, the relative contributions of biology and built environment to black–white disparities in glomerular disease have yet to be clarified. Similar observations linking *APOL1* mutations with progressive CKD or ESKD among other ancestral groups have not been described and specific “genetic” causes for the substantially higher risk of ESKD in these groups relative to European whites remain unclear.⁹⁸ Please refer to Chapters 19 and 43 for further discussion of the complex relations among ancestry, genes, and related diseases of the kidney.

HEALTH SYSTEM AND DISPARITIES IN NEPHROLOGY

More than 30 years ago, Donabedian proposed a model for assessing the quality of medical care based on structure, process, and outcome. According to the Donabedian model, “process—the method by which health care is provided—is limited by the structure or environment in which it operates.”¹¹⁰ The health system plays an important role in mediating disparities in nephrology. Health systems are responsible for promoting and implementing interventions to improve health by altering exposure to key intermediaries. However, in many countries, differential access to the health system and differential access to high-quality, integrated health care often lead to relative differences in the receipt of key interventions to promote and maintain health.

UNEQUAL HEALTH SYSTEM ACCESS AND SURVEILLANCE

Disparities in the surveillance and treatment of CKD risk factors may contribute to the higher incidence of more advanced CKD in persons of low socioeconomic status than in those of intermediate or high socioeconomic status. In the United States, more than a third of patients with incident ESKD are persons of severely limited economic means (enrolled in Medicaid, the US joint federal and state health insurance program for the poor, or uninsured at ESKD onset).² However, persons with CKD from impoverished communities are difficult to identify because there is no national system

for tracking the care of poor or uninsured patients in most countries.^{2,5,23} In other words, disadvantaged patients with nondialysis-dependent CKD are essentially “invisible” to much of the health-care system until they actually have ESKD. In many instances, CKD can be slowed or prevented. Interventions such as blood pressure lowering, angiotensin inhibitor use, and avoidance of nephrotoxins are effective in reducing morbidity and slowing progression of most kidney diseases.^{111–115} Because of differences in health system access according to socioeconomic position, however, the time and resources needed to effectively implement these interventions in patients from socially disadvantaged groups are limited, as evidenced by underutilization of these therapies among the poor and uninsured.^{12,116}

Using national data, we previously reported that in comparison with US adults with earlier stages of CKD, those with more advanced CKD were younger and more likely to be nonwhite, poor, and uninsured.¹² Consistent with the differential exposure framework from Fig. 83.1, uninsured adults with CKD had a higher prevalence of risk factors for progression to ESKD than their insured counterparts but were far less likely to receive interventions based on and recommended by clinical practice guidelines to slow disease progression.¹² Recent data from the US indicate that the scope of basic health insurance coverage likely plays a critical role in the treatment of modifiable risk factors for kidney diseases. In states with broader as compared with narrower health insurance coverage for the poor, Kurella-Tamura et al.¹¹⁷ observed a significantly lower incidence of ESKD and smaller health insurance–related gaps in care access. Such evidence highlights the potential impact of (including the legislative uncertainty around) the Affordable Care Act, which expanded eligibility criteria and broadened the range of tools and strategies available to states to provide health care for the poor and uninsured.^{118–120}

As noted earlier, prior studies showed that lack of health insurance coverage, residential poverty, and nonwhite race–ethnicity were independently associated with worse metabolic abnormalities at dialysis initiation, including more severe hypoalbuminemia, azotemia, and anemia, and with marked delays in accessing the transplant waiting list and in receiving a kidney transplant.^{33,34,121} These reports suggest that socially disadvantaged groups may receive less and perhaps substandard care in earlier stages of CKD.^{63,122} Although differences in individual-level factors such as adherence to screening and treatment recommendations may partly contribute to these disparities, it is likely that differential access to the health system plays a central role in determining which groups receive these interventions, particularly at earlier stages of CKD.¹²² Yet to a certain degree, such disparities in ESKD incidence by socioeconomic status appear to exist even within high-income countries with universal access to care.^{6,123}

IMPORTANCE OF HEALTH SYSTEM STRUCTURE

The medical care of most poor and underserved populations is concentrated in the hands of relatively few providers.¹²⁴ For example, owing to their relatively limited options for ongoing ambulatory care, many of America’s poor and underinsured seek ambulatory care from public hospitals and safety net health-care clinics.^{125,126} Even within universal health-care systems, the isolated nature of some minority

communities strongly influences the frequency and quality of care received.^{14,15,17} Several United States–based studies have now demonstrated that socioeconomic and racial–ethnic disparities in care may be partly explained by differences in where and by whom patients are treated, leading to systematic differences in the care environment, provider quality, and access to care.^{127–129} For example, Bach and associates¹²⁷ found that black and white Medicare beneficiaries were treated for the most part by different physicians. Strikingly, 80% of all outpatient primary-care visits by black patients in this study were accounted for by only 22% of physicians. Moreover, the primary-care physicians who cared primarily for black patients were less likely to be board certified, more likely to be in training, and more likely to report encountering system barriers to providing high-quality care to their patients (e.g., securing referrals to specialists). Varkey and colleagues¹²⁸ further found that clinics serving higher proportions of patients from racial–ethnic minority groups were more chaotic than and had different organizational characteristics from those of other clinics. Recent qualitative studies of nephrology and primary-care providers have identified system-level factors such as referral process delays, insufficient time to build patient partnerships, and poorly integrated nephrology and primary care as major barriers to optimal preparation for renal replacement therapy in diverse health-care settings.^{130,131} Such systematic barriers may also account for the historically lower utilization of home dialysis and preemptive kidney transplantation among black and Hispanic patients than among white patients in the US.^{132,133} In short, several key factors linked to the provision of high-quality medical care, such as coordination of care, subspecialist access, and provider experience differ markedly according to the health system. Such health system–level determinants may delay or restrict the access of poor and underserved patients to receiving adequate management of common kidney diseases and their antecedent risk factors.

DISPARITIES IN THE TREATMENT OF CHRONIC KIDNEY DISEASES

Blood pressure control is a central target to reduce morbidity and mortality in CKD.¹¹⁴ Hypertension as a contributing cause of ESKD is blamed for more than a quarter of incident cases in the United States.² Hypertension is also the most common modifiable risk factor for disease progression among uninsured American adults with CKD, affecting approximately 57% of this population as compared with obesity (40%), diabetes (22%), and overt albuminuria (13%).¹² However, several United States–based studies have described marked disparities in blood pressure control in CKD according to measures of socioeconomic status and health system access. For example, data from the US National Health and Nutrition Examination Surveys indicated that hypertensive adults with reduced eGFR (15–59 mL/min/1.73 m²) and/or albuminuria (≥30 mg/g) who lacked health insurance were far less likely to be receiving treatment for their hypertension than insured counterparts.¹² Moreover, among hypertensive adults with reduced eGFR and/or albuminuria, those who lacked health insurance coverage were 55% less likely to be receiving treatment with an angiotensin inhibitor than insured adults (Table 83.1).¹² Similarly, Olomu and associates¹¹⁶ reported

Table 83.1 Odds Ratio (OR) of Hypertension Treatment and Renin–Angiotensin Antagonist Use Among Uninsured as Compared With Insured US Adults With Chronic Kidney Diseases and Hypertension

Outcome	Unadjusted OR (95% CI)	Adjusted OR (95% CI)
Hypertension treatment ^a	0.44 (0.31–0.62)	0.59 (0.40–0.85)
Renin–angiotensin–aldosterone inhibitor use	0.34 (0.20–0.59)	0.45 (0.26–0.77)

^aTreated: We considered participants with hypertension to be “treated” if they reported that they were currently taking medications to lower blood pressure.
CI, Confidence interval.

that only 38% of hypertensive patients achieved target blood pressure control in a federally qualified health clinic in Michigan; the US average is 54%.¹³⁴ Even fewer (31%) hypertensive patients with diabetes achieved target blood pressure control in the same clinic.¹¹⁶

In addition to experiencing reduced access to high-quality care, low-income and racial–ethnic minority groups may harbor risk factors that heighten their susceptibility to further morbidity and mortality. While observing a high (10%–35%) prevalence of moderately or severely reduced eGFR (<60 mL/min/1.73 m²) among ambulatory users of a large safety net health system, we found that the prevalence of “socially determined” risk factors that might influence medical outcomes—such as chronic viral diseases, substance abuse, homelessness, and mental illness—was far higher than observed in other health-care settings.^{97,135} In one safety net setting, >7% of persons with moderate-to-advanced CKD progressed to ESKD and 16% died during median follow-up of 6.6 years.⁹⁷ Notably, nearly 30% of patients were never assessed for proteinuria, and fewer than 20% were ever seen by nephrology specialists during follow-up.⁹⁷ The elusive nature of this population, their frequent loss to follow-up, reduced access to specialty care, and limited provider accountability further complicate patient–provider interactions, highlighting a critical need for more effective strategies to optimize surveillance and management of common risk factors for kidney diseases in traditionally underserved populations in order to mitigate health status and health-care disparities.

DISPARITIES IN THE TREATMENT OF END-STAGE KIDNEY DISEASE

Conservative estimates suggest that ESKD affects at least 4.9 million people worldwide at considerable cost to society.³ In some high-income countries with universal access to ESKD treatment, there are persistent racial–ethnic and socioeconomic disparities in access to certain types of renal replacement therapies.^{2,3} In most medium- and low-income nations, access to ESKD treatment remains limited and is governed predominantly by socioeconomic position and occupation.^{3,5}

DIALYSIS: PROCESS AND OUTCOME

In terms of access to maintenance dialysis, socioeconomic disparities are evident at several levels, from the country's health region and system to the individual patient. Regular access to maintenance dialysis remains, for the most part, limited to patients receiving care in high-income and upper-middle-income nations.^{2,3,5} In most low-income and lower-middle-income countries, the majority of patients who initiate dialysis either die or stop treatment within the first 3 months because of cost restraints.^{5,23} While the numbers who do not receive treatment for ESKD is unknown, estimates suggest that at least 2.2 million people might have died prematurely in 2010 because renal replacement therapy could not be accessed, with the largest treatment gaps occurring in low-income countries in Asia and Africa.³ Within high-income nations, an increasing number of studies have focused on differences in dialysis process measures as indicators of the quality of dialysis care received.^{133,136-139} A large fraction of these studies emanate from the United States, likely reflecting incorporation of specific measures for anemia management and dialysis adequacy into facility performance assessment and care reimbursement.^{138,140}

As noted previously, there are prominent and persistent racial-ethnic disparities in process measures at dialysis initiation that largely reflect the presence and quality of care leading up to ESKD.^{33,139,141} However, data from the US Centers for Medicare and Medicaid Services (CMS) suggest that these disparities attenuate and may even reverse for some racial-ethnic groups over time.^{142,143} In two studies from the CMS's ESRD Clinical Performance Measures Project, Frankenfield and associates reported that Hispanic and Asian patients undergoing hemodialysis experienced process measures (e.g., arteriovenous fistula use and serum albumin, hemoglobin, and dialysis adequacy targets) that were equivalent, and in some cases superior, to those of non-Hispanic whites 1 year after dialysis initiation; and that black patients receiving hemodialysis continued to experience the worst process measures of all racial-ethnic groups surveyed.^{136,142} While the first study did not elucidate specific mechanisms by which racial-ethnic differences in process measures abated, it seems plausible that the establishment of a regular site of care with access to wraparound services (i.e., social work, dietician, primary nephrology care) may have offset the lack of, or variation in, medical care received prior to ESKD onset. Whether achievement of specified targets for these measures actually reflects dialysis-related quality of care remains a source of substantial debate.^{133,138} Recent voices have called for additional metrics such as kidney transplant referral rates to provide further insight into the overall quality of dialysis care delivered.¹⁴⁴ Regardless, in the setting of universal access to dialysis services, more research is needed to determine how quality improvement methods can be effectively targeted to reduce health disparities without misprioritizing low-value care and creating unwarranted and wasteful incentives that might also exclude vulnerable patients.^{133,145}

KIDNEY TRANSPLANTATION: PROCESS

In terms of survival and quality of life, kidney transplantation is the optimal treatment for most patients who progress to ESKD. However, significant disparities in receiving a transplant

according to socioeconomic status, race-ethnicity, and geography persist worldwide.^{2,11,13,146-148} In the United States, transplantation rates are significantly lower among the poor and among most racial-ethnic minority groups than among less disadvantaged groups and whites, respectively.² Racial-ethnic disparities are also present in Canada despite uniform health insurance coverage, where kidney transplantation rates in Aboriginal, African, Indo Asian, and East Asian Canadians range from one-half to two-thirds those of white Canadians.¹⁴⁶ In Australia, transplantation rates in Aboriginal Australians are 77% lower than those in nonindigenous Australians, and in New Zealand, Maori/Pacific Islanders are similarly disadvantaged.⁸ Although key mediators of, and their relative contributions to, racial-ethnic disparities in transplant access vary by population and health system, many are directly linked to social conditions described previously.

For example, in the United States, American Indian and black patients encounter delays in accessing the kidney transplant waiting list that are partly attributable to lower socioeconomic status and a higher prevalence of diabetes mellitus in comparison with white patients.^{19,34,149,150} Similarly, contextual poverty and health insurance coverage account for the largest fraction of the disparity in live-donor kidney transplantations among American Indians, Alaska Natives, and black and Hispanic Americans in comparison with white Americans.¹⁴⁸ By contrast, geographic differences (primarily attributed to organ availability) account for substantial fractions of the disparity in receiving a deceased-donor kidney from the waiting list among US Hispanics, Pacific Islanders, and American Indians/Alaska Natives.³⁴

Additional studies have identified differences in attitudes and knowledge about kidney transplantation among black and Hispanic Americans in comparison with white Americans.¹⁵¹ Providers may also be less likely to recommend kidney transplantation to minority patients owing to impaired communication and misconceptions regarding its benefits relative to prolonged dialysis.^{151,152} Emerging evidence further suggests that the dialysis center may play an influential role in promoting and facilitating access to kidney transplant, particularly among racial-ethnic minority patients.^{153,154} The influence of cultural and/or linguistic isolation on delays in accessing the transplant waiting list or in receiving a kidney transplant remains understudied.³⁴

Specific mechanisms by which social disadvantage influences access to transplantation include low educational attainment, suboptimal health insurance coverage, and contextual impediments to healthy living. Poor or uninsured patients appear to encounter more difficulty than less disadvantaged patients in navigating the complex steps required to successfully receive a kidney transplant, and this vulnerability appears to have changed little over time.^{149,155,156} This may be a direct effect of low educational attainment, inadequate insurance coverage, suboptimal patient-provider communication, and lack of support and information about transplant from dialysis centers, all of which have been linked with delays in accessing the transplant waiting list, in completing the transplant evaluation, and in successful receipt of a transplant.^{100,146,149,150,152,157,158}

Nephrology providers also appear reticent to refer poor patients for transplant because of concerns about inadequate coverage for prescription and procedural costs.^{152,159} In a national survey of US transplant programs, >70% responded that they frequently or occasionally exclude patients from

the kidney transplant waiting list owing to concerns that the patients will not be able to afford the immunosuppressive medications.¹⁶⁰

Although socioeconomic status and contextual poverty are linked with disparities in transplantation rates in the United States,^{30,78,149} the magnitude of such disparities is markedly higher in middle-income nations such as in Mexico, where transplantation rates are more than tenfold higher among insured patients than among uninsured patients.¹⁶¹ As concerning is the unethical practice of organ trafficking and exploitation of disadvantaged populations for profit that occurs in many low- and middle-income nations.¹⁶¹ This pervasive commercialization of live donor transplantation is supported by reports of affluent foreigners undergoing large fractions of the kidney transplantations performed in some low- and middle-income countries.¹⁶² Although some writers justify paying persons to donate a kidney as mutually benefiting recipient and donor, one study conducted in India showed that such transactions do not lead to long-term economic benefit and may be associated with a decline in health for the donor.¹⁶³

Historically, racial-ethnic differences in the distribution of ABO blood type and antibodies to human leukocyte antigens (HLAs) have been linked with disparities in receiving a deceased-donor kidney transplant among patients who have successfully accessed the waiting list.^{34,164,165} In particular, black candidates have encountered marked delays in receiving a deceased-donor kidney owing partly to their higher frequencies of ABO types that are associated with longer waits.^{10,22,165–167} Such factors may account for marked delays in kidney transplantation among black as compared with nonblack children, which persisted even when accounting for socioeconomic measures.⁹¹ However, the influence of biologic perpetrators of racial-ethnic disparities in transplantation access, such as the distribution of blood type and HLA, has progressively declined as a result of major changes in the deceased donor allocation policy. Recognizing the contribution of HLA matching to racial disparities in receiving a deceased-donor kidney, US policy makers eliminated the HLA-B locus as a priority for the allocation of deceased kidney donors on May 7, 2003.¹⁶⁸ This change led to an increase in the proportion of deceased-donor organs directed to minority candidates on the waiting list. Specifically, in the 6 years after the policy change, deceased-donor transplantation in racial-ethnic minority recipients rose 40%, compared with an 8% rise for non-Hispanic whites and a 23% increase overall.¹⁶⁹ Moreover, this improved access to transplantation for minority groups has not been accompanied by declines in short-term graft survival during the initial period after the policy change.¹⁶⁹

In 2013, the US Organ Procurement and Transplantation Network approved additional amendments to the kidney allocation policy that included changes to promote greater equity in accessing deceased-donor kidneys.¹⁶⁶ In particular, there were two amendments, one that allows for blood type B recipients to receive kidney offers from donors with certain subtypes of blood type A and another that calculates waiting time from the start of dialysis (or qualifying eGFR ≤ 20 mL/min). In the first 9 months under the new allocation system, incidence rates of deceased-donor kidney transplantation improved by 13% and 19% for Hispanic and black patients (as compared with rates under the old allocation system),

respectively, but these improvements in transplant access for racial-ethnic minorities were tempered by higher rates of delayed graft function.^{166,170,171}

Over the past decade, kidney exchange has gained substantial momentum in the US and has prompted forward thinking on how the global nephrology community might leverage this approach to mitigate racial-ethnic and socioeconomic inequities in kidney transplantation. Additional strategies to promote greater racial-ethnic and social equity in rates of assigning patients to the waiting list and in living-donor kidney transplantation remain areas of heightened investigation.¹⁵⁴

KIDNEY TRANSPLANTATION: OUTCOMES

Socioeconomic and racial-ethnic disparities in kidney transplantation extend to outcomes after receipt of a functional allograft. Similar to transplant access, specific causes of these disparities in allograft survival may differ by population and health system, but the root causes commonly reflect differential social conditions and access to high-quality health care. For example, adherence to posttransplantation immunosuppressive therapy remains a critical determinant of long-term allograft survival. Regrettably, the inability to afford immunosuppressive medications appears to be a primary mechanism that restricts disadvantaged patients from not only receiving a kidney transplant but also in maintaining one.^{170,172} In the United States, for example, federal prescription drug coverage for immunosuppressive medications extends to only 36 months after kidney transplantation, with few exceptions.¹⁷³ For this reason, disadvantaged patients are particularly vulnerable to the long-term financial burden of maintaining a functional allograft.¹⁷⁴ In one study, >40% of transplant recipients reported financial difficulty after transplantation, with health-related out-of-pocket expenses averaging approximately \$475 per month.¹⁷² In a recent single-center study of 1171 kidney transplant recipients, lower socioeconomic status was associated with an 11% higher risk of acute rejection and 77% higher risk of graft loss among black Americans.¹⁷⁵ Socioeconomic disparities in maintaining a transplant are magnified in medium-income countries such as Mexico and Pakistan, where excellent long-term results for allograft survival are offset by the prohibitive cost of immunosuppressive and antiviral medications. In these countries, transplant access and allograft maintenance for the average citizen often require a state-sponsored model.^{5,47,161}

In terms of the disparity in allograft loss between black and white Americans, the gap has steadily declined over the past 2 decades. Recent data from the US organ transplant registry point to differential improvements in 5-year graft survival after deceased-donor kidney transplantation among black (from 49% to 69%) as compared with white (from 63% to 75%) Americans. Similar gains (63% to 79% among blacks vs. 79% to 86% among whites) have led to a narrowing of the black-white gap in allograft survival following living-donor kidney transplantation.¹⁷⁶ Temporal changes in deceased donor allocation, immunosuppressive agents, and desensitization strategies have been credited as major contributors to these observations.

Racial differences in biologic factors such as age at transplantation, HLA mismatching, and APOL1 genotype status may account to some degree for the remaining disparity in

allograft survival between some racial-ethnic groups, such as between black and white Americans.^{10,177,178} The transplant community is now seeking ways to integrate the APOL1 genotype into clinical practice in order to enhance evaluation of donor organs and better align this assessment with recipient need.¹⁷⁹ Although prevailing beliefs attribute reduced graft survival among black recipients to heightened immune responsiveness and poorer immunologic matching, remarkably few reports have actually linked better HLA matching with improved graft survival in this racial group.^{10,164,180} In addition, Asian recipients enjoy the longest graft survival of all racial-ethnic groups despite the presence of suboptimal HLA matching comparable with that observed in black recipients.^{10,181} Alternatively, pharmacokinetic studies report greater clearance and/or lower bioavailability of certain immunosuppressive agents among black than among white transplant recipients.^{182–184} However, the influence of these biologic factors on racial disparities in allograft survival is likely mediated at least in part through differences in immunosuppressive medication regimens and nonadherence.^{173,185–187} In the United States, nonadherence strongly correlates in turn with social factors such as household income, presence and type of health insurance coverage, and history of drug abuse.^{10,173,188,189}

Lastly, studies have raised concerns regarding inequities in the quality of transplanted organs based on recipient socioeconomic status and race-ethnicity.¹⁹⁰ Emerging data on provision of comparatively “lower quality organs” to black, low-income, or less educated recipients than to white, more affluent, or college-educated counterparts have cast additional light on the complex nuances that underlie this social gap in allograft access and survival.^{190,191}

MOVING TOWARD MORE EQUITABLE CARE

Currently, global cost estimates of treating kidney diseases remain elusive. In the United States alone, treatment of CKD costs the federal government >\$50 billion annually, including \$33 billion for ESKD.² Hence, there is an urgent need to identify cost-effective strategies to address risk factors for major kidney diseases, particularly among populations that bear a disproportionate burden of disease.² Strategies to reduce disparities in chronic diseases include three general approaches: (1) targeted programs, (2) gap programs, and (3) gradient programs.^{68,192} “Targeted programs” comprise strategies aimed at improving the health of groups that are particularly disadvantaged in terms of disease burden or treatment access. Such targeted programs are advantageous in that they focus on a well-defined, often small segment of the population. This aspect provides for ease in monitoring and assessing results. However, the targeted approach is similarly weakened by its narrow focus and its lack of commitment to reduce the disparity between the most and least disadvantaged groups. In some instances, health gains in the targeted groups may still lag behind those observed in less disadvantaged groups, leading to widening of the disparities.⁶⁸ An example is programs that attempt to promote live-donor kidney transplantation among patients of low socioeconomic status. Although such programs may increase live-donor kidney transplantation rates among the poor, the disparity between patients of lower and those of intermediate or higher

socioeconomic status may actually increase owing to the differential effects of other programs (e.g., paired donation) or resource-mediated advances (e.g., Internet use and social networking).

Strategies that aim to target the health gap address the issue of disparities, but these programs are more challenging than those that seek to improve the health status of targeted groups. In order to succeed, “gap programs” must achieve absolute improvements in health status among, for example, persons in the lowest socioeconomic position at a rate of improvement that exceeds that observed in the comparison group. Like targeted programs, gap programs often focus solely on the most disadvantaged groups and largely ignore gradients in health and the health status of those in the intermediary groups.⁶⁸ For example, over the past 14 years, collective efforts to increase kidney donation in the United States among black Americans have yielded a 30% increase in the donation rate of deceased-donor kidneys (from 18.1 per 1000 deaths in 2000 to 23.6 per 1000 deaths in 2014) from this racial group. By comparison, the donation rate from white donors remains higher (29.8 per 1000 deaths in 2014), but this rate has been relatively stagnant over the same period (note: donation rates from deceased donors of American Indian and Asian descent have increased substantially during the same period, but the small number of traumatic deaths occurring in these racial-ethnic groups provides less stable estimates).²

Addressing socioeconomic gradients in health relies on comprehensive strategies that involve the entire population. In contrast to targeted and gap programs, “gradient strategies” address the effect of socioeconomic inequality on health across the socioeconomic hierarchy. In other words, a gradients model addresses not only intermediaries of health disparities among the disadvantaged but also systematic inequities in life course exposures (educational and occupational opportunities, nutrition, living standards, and health-care access) that strongly influence an individual’s position in the socioeconomic hierarchy.⁶⁸ Needless to say, gradient strategies must often contend with conflicting political agendas, major logistical challenges, and prohibitive costs.¹⁹³ Moreover, such strategies commonly yield results only in a prolonged time frame. These three strategies (targeted, gap, and gradient) for reducing disparities are intended to complement one another and can often provide sequential layers for addressing socioeconomic (and racial-ethnic) inequities.⁶⁸

On the basis of the framework shown in Fig. 83.1, researchers have identified four major intervention areas for addressing socioeconomic and racial-ethnic disparities in kidney disease.⁶⁸ First, reducing inequalities in the distribution of structural determinants such as education, occupation, and income may alter social stratification and mitigate its effects on health outcomes. Second, addressing specific intermediary determinants that mediate the effects of socioeconomic position on health (e.g., consumption of high-calorie foods and beverages, lack of physical exercise, unhealthy living conditions, and cigarette smoking) may reduce specific health-damaging exposures that disproportionately affect disadvantaged groups; in other words, changing the distribution of risk factors related to socioeconomic position may attenuate long-term health consequences for underprivileged groups.²⁸ Third, reversing the effect of health status on

socioeconomic position may lessen the vulnerability of disadvantaged groups to the health-damaging conditions they face²⁴; examples include programs to keep persons with chronic diseases such as diabetes and CKD within the workforce.¹⁷⁴ Lastly, targeting the delivery of health care to socially disadvantaged groups may reduce the unequal consequences of illness and prevent further disease progression and economic loss; such strategies might offer additional care or health coverage to members of disadvantaged groups who have conditions (obesity, diabetes, hypertension, homelessness) that are linked to the excess burden of progressive CKD in these groups.⁶⁸

INITIATIVES TO REDUCE DISPARITIES

Successful initiatives to reduce socioeconomic and racial-ethnic disparities in CKD risk factors might reference the US government-sponsored Vaccines for Children Program, the Standards of Care for Diabetes and Kidney Disease Programs administered by the US Indian Health Services, and the US National Health Service Corps' scholarship and loan repayment programs. A generation ago, fatal outbreaks of measles in predominantly poor, minority communities in several US cities underscored marked disparities in vaccination coverage that placed disadvantaged children at increased risk for disease. In direct response to these disparities, the US government initiated the Vaccines for Children Program, which now purchases and provides vaccines for children who are particularly vulnerable to lapses in vaccination coverage (i.e., those who were uninsured, on Medicaid, or of American Indian or Alaska Native ancestry, or who had health insurance that did not cover universally recommended vaccines). From 2000 to 2010, racial-ethnic disparities in vaccination series completion rates in the United States decreased from 6% to 4% (for minority groups vs. non-Hispanic white children), and poverty-related inequalities from 8% to 3%.¹⁹⁴

Several decades ago, in response to rising concerns over the burden of ESKD due to diabetes in many American Indian communities, the Indian Health Service established the Standards of Care for Diabetes Program (followed later by the Kidney Disease Program) in order to improve the screening of and the management of patients with diabetes and CKD.¹⁸ Based on the Chronic Care Model, these two programs implemented routine reporting of eGFR, annual monitoring of protein excretion, utilization of renin-angiotensin system antagonists, and aggressive control of glucose and blood pressure in association with enhanced patient and provider education (Fig. 83.2).^{18,195} By 2001, age-adjusted ESKD incidence among American Indians with diabetes had decreased by 31% from that in 1990. By 2006, 82% of American Indians with hypertension and diabetes were receiving a renin-angiotensin-aldosterone system inhibitor. Remarkably, among American Indians, age-adjusted incidence rates of diabetes-related ESKD have decreased by 54%, from 573 per million population in 1996 to 265 per million population in 2013, and are now the same as those for non-Hispanic whites (Fig. 83.3).^{2,18,196} A similar gap initiative by the Australian government, the Medical Outreach-Indigenous Chronic Disease Program, aims to close the disparity in life expectancy between indigenous and nonindigenous Australians by improving access of indigenous people to best-practice chronic disease management.²³ Notably, the five areas of highest priority include diabetes mellitus and CKDs.

In terms of addressing disparities in kidney transplantation, recent efforts have targeted dialysis facilities with substantial racial disparities in transplant referral while others have aimed to streamline the cumbersome medical work-up. The Reducing Disparities in Access to kidney Transplantation (RaDIANT) Community Study, an ESKD network-led, randomized, pragmatic trial of 134 dialysis facilities, tested the effectiveness of a 1-year multicomponent intervention (a combination of outreach, education, and quality improvement) targeting dialysis facility patients, staff, and leadership on rates of kidney

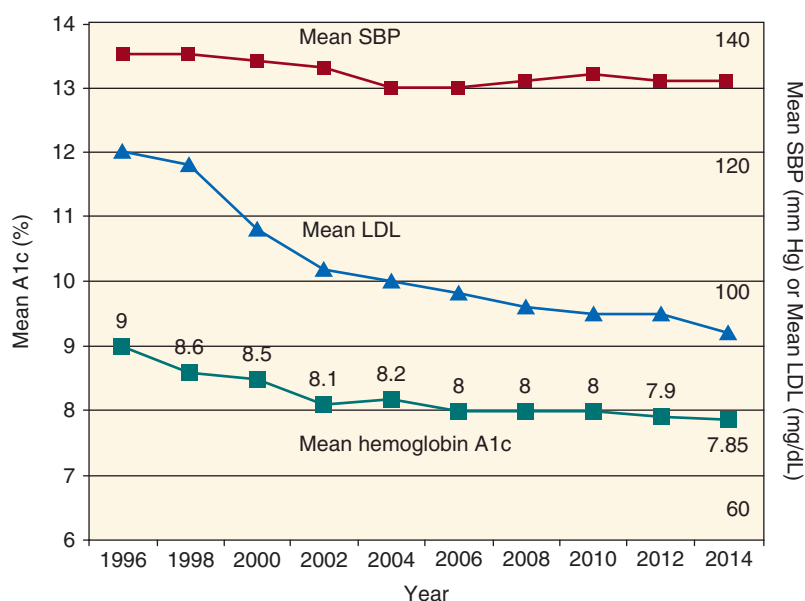


Fig. 83.2 Risk factor control among American Indians with diabetes mellitus within the Indian Health Service. LDL, Low-density lipoprotein; SBP, systolic blood pressure. (Data from Indian Health Service website at <http://info.ihs.gov/Diabetes.asp>.)

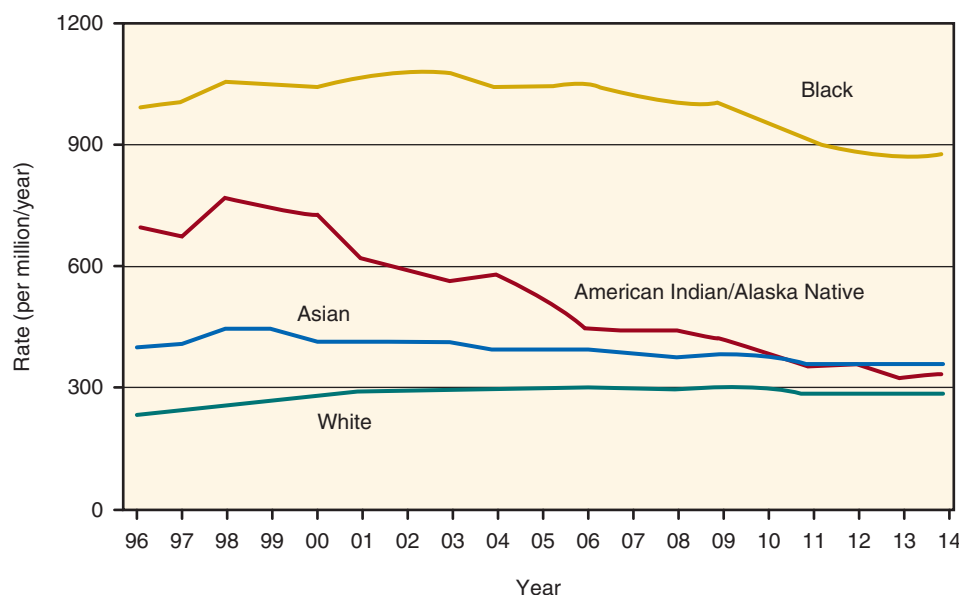


Fig. 83.3 Trends in age- and sex-adjusted incidence rate (per million/year) of end-stage kidney disease by race in the United States. (Data from US Renal Data System. *USRDS 2016 Annual Data Report: Atlas of Chronic Kidney Disease and End-Stage Renal Disease in the United States*. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2016.)

transplant referral.¹⁵⁴ After adjustment for time and clustering, the percentage of patients referred for transplant evaluation 1 year after intervention start was 7.3% (95% confidence interval, 5.5% to 9.2%) higher in intervention as compared with control facilities. Notably, the adjusted increase in transplant referrals in the intervention as compared with the control facilities was more pronounced for black (6.4%) as compared with white (3.7%) patients.¹⁵⁴ A second study, The Kidney Transplant Fast Track – Talking About Live Kidney Donation study, aims to address a major factor contributing to racial disparities in kidney transplantation, lower rates of live-donor kidney transplantation. This study will test the effectiveness of a 1-day, streamlined evaluation process combined with a randomized controlled trial of the Talking About Live Kidney Donation intervention, a validated, culturally sensitive program to support consideration of live-donor kidney transplantation among patients and their families.^{197–199}

In 1970, the US government established the National Health Service Corps to help underserved communities across the nation receive critically needed primary medical, oral, and mental and behavioral health care. Through the National Health Service Corps, clinicians receive scholarships and loan repayment in return for committing to practice in underserved areas for a defined period. More than 40,000 primary-care clinicians have participated in this program since its inception. Later reports suggest that 82% of clinicians continue to practice in underserved communities 1 year after their service completion and that 55% remain for 10 years. Notably, long-term retention rates are higher for those who serve in rural as opposed to urban communities.²⁰⁰

Many US policy makers and health-care providers lauded passage of the Affordable Care Act as a fundamental step toward achieving equitable access to health care for all Americans. However, uncertainty surrounding its legislative future continues to shroud robust gains in health-care coverage since enactment.¹²⁰ Furthermore, as Schrag²⁰¹ noted in

1983, the ability to pay for health services does not necessarily guarantee their availability. In order to address socioeconomic and racial-ethnic disparities in treatment of kidney disease and its antecedent risk factors, US policy makers will need to work with health systems to tackle the unequal distribution of resources including experienced health-care providers situated in areas of low and high poverty. In terms of kidney transplant, efforts to reduce economic vulnerability by providing medication prescription benefits to disadvantaged groups at risk for nonadherence to posttransplantation regimens may partially address socioeconomic and racial-ethnic inequities in transplant access as well as in long-term allograft survival.¹⁷³ To date, lobbying efforts have failed to convince the US Congress to extend support for immunosuppressive medications beyond 3 years after transplantation.

SUMMARY AND CHALLENGES

Socioeconomic and racial-ethnic disparities in the incidence and treatment of kidney diseases and their risk factors occur worldwide. They occur within and between countries, and the reader is referred to the Global Considerations in Kidney Disease section of this edition (Chapters 75 through 81) for consideration of particular constituencies. Growing recognition of these disparities has fueled additional research and programs aimed at addressing gaps between persons of low socioeconomic status and their less impoverished counterparts, and between racial-ethnic minority groups and white majority groups. However, despite in-depth knowledge of the root causes and consequences of these disparities, including their social determinants, progress remains slow and the structural elements that allow these disparities persist. Marked differences in education, living conditions, and employment opportunities combined with inequitable access to high-quality health care and genetic susceptibility act in tandem to promote

and maintain disparate milieus for chronic diseases. In the setting of such widespread inequities, we should anticipate that disparities in health outcomes such as kidney diseases will continue to plague the nephrology community for the foreseeable future. The persistence of, and costs attributed to, potentially preventable disparities in nephrology mark our field as one that remains in crisis.

Fortunately, several countries including the United States have now prioritized initiatives aimed to reduce socioeconomic gaps in health insurance coverage, maternal prenatal care, vaccinations, smoking, nutrition, and childhood obesity in order to improve the overall quality of health care to, and mitigate the effects of social conditions that disproportionately afflict, socially disadvantaged populations. Within nephrology, targeted efforts over the past 2 decades have yielded substantial declines in diabetes-related ESKD incidence among American Indians and improved access to deceased-donor kidneys among black and Hispanic Americans. However, for the long term, reducing disparities rooted in social inequity will likely improve health and health-care quality more than would technical advances in overall medical care.¹⁹⁴ The social distribution of prematurity, low birth weight, diabetes mellitus, hypertension, and obesity and their indelible link to progressive declines in kidney function provides further evidence in favor of this approach to address disparities in nephrology.²⁸ Unfortunately, many high- and medium-income countries continue to witness stagnant or reduced social mobility, increased social unrest, and widening inequities in living conditions. Needless to say, even in the presence of astute public policies aimed to address marked differences in social conditions, reducing long-standing disparities in nephrology will remain a major challenge for the foreseeable future.



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BOARD REVIEW QUESTIONS

- Worldwide, incidence rates of end-stage kidney disease (ESKD; mainly diabetic kidney disease) remain higher among most indigenous groups as compared with white majority groups. In which of the following indigenous groups has the incidence of diabetes-related ESKD steadily declined to the point that it is now comparable with the corresponding white majority group?
 - Pacific Islander Americans
 - American Indians
 - Aboriginal Australians
 - Aboriginal Canadians
 - None of the above

Answer: b

Rationale: Several decades ago, in response to rising concerns over the burden of ESKD due to diabetes in many American Indian communities, the Indian Health Service established the Standards of Care for Diabetes Program (followed later by the Kidney Disease Program) in order to improve the screening of and the management of patients with diabetes and chronic kidney disease. By 2006, 82% of American Indians with hypertension and diabetes were receiving a renin–angiotensin–aldosterone system inhibitor. Notably, age-adjusted incidence rates of diabetes-related ESKD has decreased 54% among American Indians, from 573 per million population in 1996 to 265 per million population in 2013, and is now the same as that for non-Hispanic white Americans. Incidence rates of diabetes-related ESKD remain substantially higher among the other indigenous groups relative to white majority groups.

- Several key factors linked to the provision of high-quality medical care differ markedly according to the health system. Such health system–level determinants may delay or restrict the access of poor and underserved patients to receiving adequate management of common kidney diseases and their antecedent risk factors. Which of the following factors is **UNLIKELY** to contribute to systematic differences in the delivery of care to traditionally underserved groups (as compared with less disadvantaged groups)?
 - Coordination of specialty and primary care
 - Subspecialist access
 - Provider experience
 - Adequate time to build patient–provider partnership
 - None of the above

Answer: e

Rationale: Recent US-based qualitative studies of nephrology and primary-care providers have identified system-level factors such as referral process delays, insufficient time to build patient partnerships, and poorly integrated nephrology and primary care as major barriers to optimal preparation for renal replacement therapy in diverse health-care settings. Prior studies have also identified differences in provider training and qualifications in clinics that primarily serve poor or underserved groups as compared with clinics that serve less disadvantaged populations.

- In terms of reducing socioeconomic disparities in nephrology, which of the following interventions might prove most effective?
 - A policy-level intervention aimed at reducing inequities in primary and secondary education
 - A multifaceted intervention that promotes healthy living in disadvantaged communities, eg, reduces consumption

- tion of high-calorie foods and beverages, increases exercise, addresses unhealthy living conditions)
- An intervention program that offers additional care or health coverage to members of disadvantaged groups
- An intervention program aimed at keeping persons with diabetes mellitus and/or advanced chronic kidney disease in the workforce
- All of the above
- None of the above

Answer: a

Rationale: The strategies listed above cover the four major intervention areas for addressing socioeconomic disparities in health care. However, addressing fundamental inequities in education are likely to provide the greatest benefits. The other interventions lie downstream from this social (or structural) determinant.

- Disparities in the incidence of end-stage kidney disease arise through various pathways. Which of the following actions is least likely to influence socioeconomic disparities in the incidence or treatment of chronic diseases of the kidney?
 - Expansion of health insurance coverage for low-income individuals
 - Enhanced health surveillance for underserved populations
 - Universal access to prenatal care
 - Issuance of government housing vouchers allowing relocation to less distressed neighborhoods
 - All of the above entities will influence socioeconomic disparities in the incidence or treatment of major chronic diseases of the kidney

Answer: e

Rationale: All of the above entities are likely to influence chronic kidney disease (CKD) incidence or treatment through pathways discussed in the chapter (nephron endowment, access to health-care facilities or services, social or behaviorally patterned antecedents of CKD). Recently, significantly lower incidence of end-stage kidney disease and smaller health insurance–related gaps in care access were observed in US states with broader as compared with narrower health insurance coverage for the poor.

- Strategies to reduce disparities in chronic diseases include three general approaches: (1) targeted programs, (2) gap programs, and (3) gradient programs. Targeted programs comprise strategies aimed at improving the health of groups that are particularly disadvantaged in terms of disease burden or treatment access. In terms of reducing health disparities among the targeted (least advantaged) and most advantaged groups, potential drawbacks to the targeted program approach include which of the following?
 - Higher costs than gap or gradient programs
 - Inability to account for health gains in the most advantaged groups
 - Inability to address health status of intermediary groups
 - Difficulty in identifying and tracking target population
 - All of the above
 - None of the above

Answer: b

Rationale: Technically c is also a correct statement, but it is irrelevant in terms of reducing health disparities among